
CSE 151B Final Report — Team Name

Ketan Mittal

Department of Data Science
University of California, San Diego
k1mittal@ucsd.edu

Anirudh Annabathula

Department of Data Science
University of California, San Diego
aannabathula@ucsd.edu

Kaii Bijlani

Department of Data Science
University of California, San Diego
kbijlani@ucsd.edu

Abstract

This is our final report for the CSE 151B Spring 2026 Math Reasoning Competition. The data is 1,126 public math problems (375 MCQ, 751 free-form) and a 943-item private test set. Kaggle has two boards: the public board scores on all of `private.jsonl`, and the private board scores on a separate hidden subset (revealed at the deadline) that catches overfitting on `private.jsonl`. The milestone got us to 0.515 public / 0.459 private with a grader-aware system prompt.

For the final phase we added three methods on top of that prompt: a 24-config sampling-parameter sweep, self-consistency majority voting over k rollouts, and GRPO LoRA fine-tuning of Qwen3-4B-Thinking-2507 on a Claude-filtered subset of the public set. We also moved inference from INT8 on one GPU to BF16 with vLLM tensor-parallel = 2 on 2× A40, which roughly quadrupled throughput. The final submission scored **0.699** on the public board (rank 42/105) and **0.627** on the private board (rank 27/105), up +0.184/+0.168 absolute and +36%/+37% relative from the milestone.

The sampling sweep was the biggest gain, and self-consistency and GRPO were both flat on our 44-item eval slice — we kept them for robustness rather than measured gains. Free-form accuracy sat near 65% across every method we tried, and that is the bottleneck we did not move. The LoRA adapter is public on HuggingFace at [k1mittal/cse151b-grpo-lora](https://huggingface.co/k1mittal/cse151b-grpo-lora), and the whole pipeline runs from `run_inference()` in `run_inference.py`. Code is at github.com/km322/CSE_151B_Final_Project.

1 Introduction

Problem Definition. Each test item is either a multiple-choice question (MCQ) with up to ten labeled options, or a free-form question that wants one or more numeric or symbolic values. The model gets the question text (and the option list when there is one), and has to produce a final answer wrapped in a boxed expression. The provided `judger.py` then parses and scores that answer. For free-form items the judger uses a sympy-based comparator with a 10^{-8} relative tolerance; for MCQ items it just expects a single capital letter inside the box. So the task is really a constrained generation problem: the model has to do multi-step math reasoning *and* put the final answer in a specific format the judger can parse.

Problem Significance. Math reasoning is one of the things people most want LLMs to do well — tutoring tools, science assistants, and many engineering workflows assume the model can both

compute the right answer and report it without ambiguity. The gap between a model that "gets" a problem internally and one that emits an answer the grading code will accept is a real, practical robustness problem, and that gap is exactly what this competition exposes.

Technical Challenge. This project is harder than it looks for a few reasons. (1) **Data:** the public set covers algebra, calculus, statistics, probability, linear algebra, geometry, number theory, and discrete math, and many of the LaTeX strings are slightly corrupted (e.g., `frac` or `int` without backslashes). Some of the published gold answers are themselves wrong or ambiguous; we use Claude as a filter to drop those before GRPO training. (2) **Grader:** the 10^{-8} *relative* tolerance means a correct-looking decimal like 0.333 is graded wrong against gold $\frac{1}{3}$; the boxed-extraction code only keeps the last contiguous group of boxed expressions after `</think>`. Either of these can quietly cost points. (3) **Model:** 4B parameters is small for math; the model loses the thread on harder items and restates decimals mid-solution. (4) **Engineering:** the milestone ran INT8 on one GPU. For the final phase we moved to BF16 with vLLM tensor-parallel = 2 on $2 \times$ NVIDIA A40 (46 GB each). Qwen3-4B is only ≈ 8 GB in BF16, so quantization was mostly dequant overhead, and the BF16+TP= 2 move roughly quadrupled throughput. (5) **Evaluation:** a full pass over 943 private items at our final $k = 3$ setting takes about four days, so we developed methods on a 44-item eval slice as a check.

Contributions. The starter notebook `starter_code_cse151b_comp.ipynb` gives an end-to-end skeleton: uv environment, vLLM model loading, MCQ and free-form prompts, batched generation, and scoring. The first thing we added (reported in the milestone) was a grader-aware system prompt that respects the judge’s parsing rules — exact symbolic form, single boxed group — which took us to 0.515 public / 0.459 private on the Kaggle leaderboards. The rest of our work sits on top of that prompt.

- **Contribution A (milestone):** a grader-aware system prompt for both free-form and MCQ items, written around the judge’s parsing rules. Pinned for everything downstream.
- **Contribution B (final):** three method layers on top of A: a 24-configuration sampling-parameter sweep, self-consistency voting over k rollouts, and GRPO [2] LoRA fine-tuning on a Claude-filtered subset of the public set. Together with the move from INT8 on one GPU to BF16 with vLLM tensor-parallel = 2 on $2 \times$ A40, these took our Kaggle public score from 0.515 to 0.699 (rank 42/105) and our private score from 0.459 to 0.627 (rank 27/105).
- **Contribution C (final):** one reproducible entry point, `run_inference()` in `run_inference.py`, that loads the base model, downloads the LoRA adapter from `k1mittal/cse151b-grpo-lora` on HuggingFace Hub if it isn’t already local, runs the full SC pipeline on the private set, and writes the submission CSV.

2 Related Work

LLMs for mathematical reasoning. Open-weight "thinking" models like Qwen3-4B-Thinking-2507 [1] are trained to reason inside `<think>...</think>` tokens before answering. The Qwen team’s evaluations recommend specific sampling settings (`temp = 0.6`, `top-p = 0.95`, `top-k = 20`) and a trailing user instruction ("Please reason step by step, and put your final answer within `\boxed{\}`"). We had been using these at the milestone without ever checking them against alternatives.

GRPO and RL fine-tuning for math. Group Relative Policy Optimization (GRPO) was introduced in DeepSeekMath [2] as a way to fine-tune reasoning models with a verifier signal, and it showed that a deterministic grader as the reward function can noticeably improve a base model’s math accuracy. The same recipe was used in Qwen2.5-Math [3] and again in Qwen3 [1], where the team collected around 4,000 query-verifier pairs and used GRPO after a small cold-start SFT phase. We have the judge as a 0/1 reward and the same Qwen3 backbone, so the recipe carries over, though our run came back neutral rather than positive (Section 4.4).

Prompt engineering for structured output. A recurring finding in LLM-with-grader pipelines is that spelling out the grader’s parsing rules in the system prompt helps more than describing the answer’s semantics alone. Our milestone prompt is an instance of this idea, specialized to the 10^{-8} relative tolerance and the contiguous-boxed-group rule.

3 Methods

Problem Setting. We treat each item as a tuple (q, O, y) where q is the question text, O is the option list (empty for free-form items), and y is the gold answer. The model defines a conditional distribution $p_\theta(r \mid s(q, O), u(q, O))$ over response strings r . The judge $\mathcal{J}(r, y, O) \in \{0, 1\}$ pulls the last contiguous group of boxed expressions from r (only after `</think>`), normalizes the LaTeX, and compares to y with sympy at 10^{-8} relative tolerance. We maximize $\mathbb{E}_{(q, O, y) \sim \mathcal{D}} \mathbb{E}_{r \sim p_\theta} [\mathcal{J}(r, y, O)]$. At the milestone we only changed s ; the final phase keeps that prompt and also changes the sampling distribution, votes across samples, and updates θ through a LoRA adapter trained with GRPO.

For fast iteration we use a fixed 44-item eval slice: the first 50 items of `public.jsonl` minus six IDs (`{0, 1, 8, 12, 13, 14}`) whose gold answers we judged wrong or ambiguous by hand. The slice is 10 MCQ and 34 free-form. Every method below is measured on the same slice; the leaderboard is the ground truth.

Idea Summary. The milestone’s grader-aware prompt was already doing most of the format work, so we kept it and added three things on top of it for the final phase: (a) a 24-config sweep over sampling parameters; (b) self-consistency voting over k rollouts; and (c) a GRPO LoRA adapter trained on a Claude-filtered subset of the public set. Each layer targets a different failure mode — the decoding distribution, per-question variance, and the policy itself — so any one can be dropped without breaking the rest. The main risk for this design was that the layers would fail to stack: SC tends to add less once sampling is already sharp, and a single epoch of GRPO on a 4B policy might not move accuracy at all. Our fallback if GRPO came back flat was to ship swept sampler + SC at the smallest k that still gave coverage — which is what ended up happening.

Method Description. We describe each of the three layers in turn. The model still receives the raw question (and the option list for MCQ items) and emits text whose final boxed group the judge parses; we do not pre- or post-process either side beyond what the judge does itself. The sampling-sweep and self-consistency layers do not learn parameters, so they need no train/val split; the full Claude-filtered public set is the dev pool for GRPO. Figure 1 sketches the final inference pipeline.

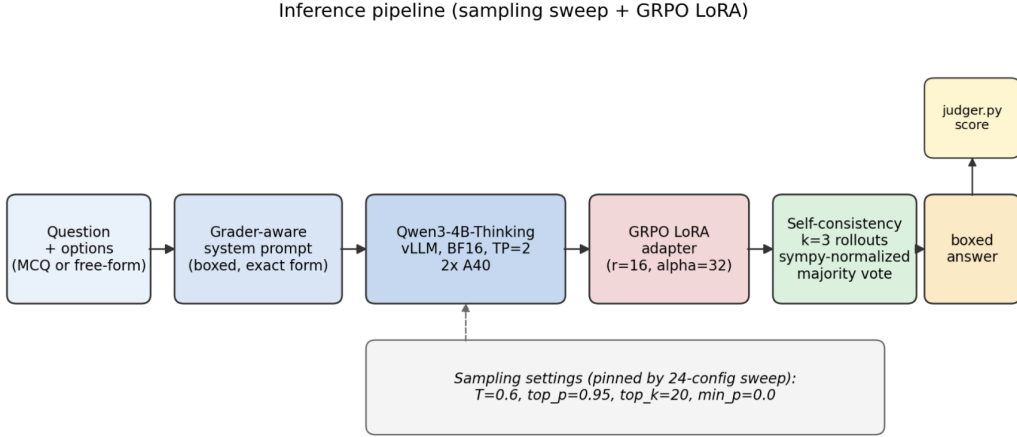


Figure 1: Final inference pipeline. The question (and option list, if any) is wrapped in the milestone’s grader-aware system prompt and decoded by Qwen3-4B-Thinking-2507 under vLLM (BF16, TP= 2 on 2× A40) with the GRPO LoRA adapter active. Sampling parameters are pinned from the 24-config sweep. We draw $k = 3$ rollouts and the sympy-normalized majority vote is scored by `judge.py`.

Sampling parameter sweep. We swept temperature $\in \{0.3, 0.6, 0.9\}$, $\text{top_p} \in \{0.8, 0.95\}$, $\text{top_k} \in \{0, 20\}$ and $\text{min_p} \in \{0.0, 0.05\}$ — 24 configurations — with `repetition_penalty=1.0` and `max_tokens=20,000`. 3 samples per question, with per-sample seeds reused across configurations so cell-to-cell deltas are not seed luck. The winner was ($T=0.6$, $\text{top_p}=0.95$, $\text{top_k}=0$, $\text{min_p}=0$) at 68.18% overall (80.0% MCQ, 64.71% free-form). The Qwen-recommended ($T=0.6$, $\text{top_p}=0.95$, $\text{top_k}=20$, $\text{min_p}=0$) tied at 68.18%. top_k barely mattered, and $\text{min_p}=0.05$

consistently hurt free-form. We pinned the recommended setting for everything after. This was the biggest gain in the final phase. Figure 2 shows the full 24-cell landscape.

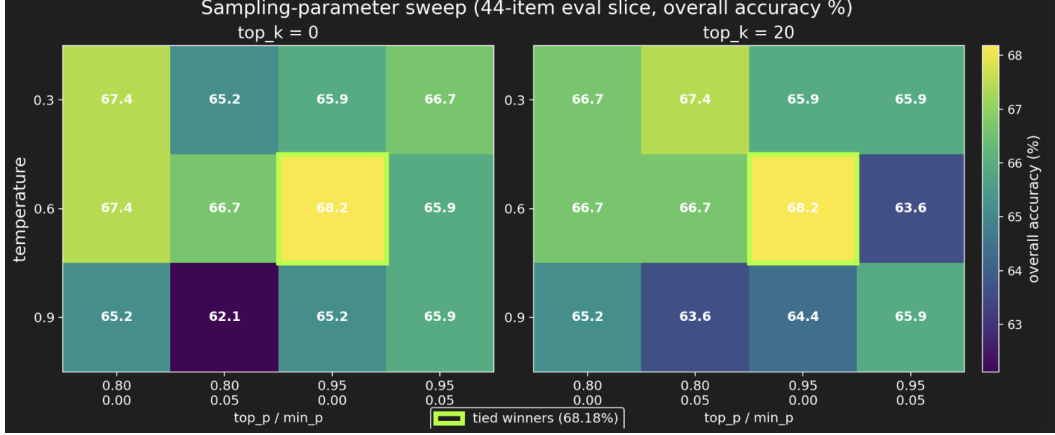


Figure 2: 24-configuration sampling sweep on the 44-item eval slice. Cells are overall accuracy (%); the two tied winners at 68.18% (both at $T=0.6$, $\text{top}_p=0.95$, $\text{min}_p=0$, with $\text{top}_k \in \{0, 20\}$) are outlined. Temperature is the dominant axis: 0.9 is consistently worst, and $\text{min}_p=0.05$ hurts the high-temp column.

Self-consistency. For each item we ran $k = 8$ independent rollouts and voted: for MCQ items the modal extracted letter; for free-form items the modal judger-normalized symbolic form, so that $\frac{1}{2}$, $\frac{1}{2}$, and 0.5 collapse into one vote. Ties broke by first occurrence. On the eval slice the vote scored 90.0 / 64.71 / 70.45 (MCQ / free-form / overall): on free-form the modal answer matched a single rollout on every item, and on MCQ one item out of ten flipped versus the best single sample (within noise on a 10-item split). Items the model got right were already high-confidence; items it got wrong tended to agree on the same wrong answer. We dropped k from 8 to 3 for the final private run — about $2.7\times$ cheaper at the same expected accuracy on this slice.

GRPO LoRA fine-tuning. We tried GRPO [2] over a Claude-filtered subset of the public set (1,113 of 1,126 items, dropping the ones Claude flagged as wrong or ambiguous). Policy was Qwen3-4B-Thinking-2507 loaded in 4-bit nf4 with a LoRA adapter (rank 16, $\alpha = 32$, dropout 0, target modules `q_proj`, `k_proj`, `v_proj`, `o_proj`). The trainer was trl’s `GRPOTrainer` launched via `accelerate launch -num_processes=2` (DDP, one rank per GPU with `device_map` pinned to local rank, colocated vLLM rollouts). The reward was a wrapper around `Judger.auto_judge` (and an exact letter match for MCQ): +1 correct, 0 otherwise. Hyperparameters: learning rate 1×10^{-6} , KL $\beta = 0.04$, `num_generations=4` (we tried 8 and ran out of memory), `max_completion_length=2,048`, per-device batch size 1 with gradient accumulation 4, one epoch (561 steps, ≈ 14 hours on $2 \times A40$). On the eval slice the GRPO adapter scored 70.45%, identical to base + SC. Probable causes: one low-LR epoch, group size 4 instead of the 8–16 used in the published recipes, completions truncated at 2,048 tokens, and a sparse 0/1 reward. We kept the adapter in the final pipeline because validation showed no regression.

Two engineering issues stalled training. A version mismatch between `peft` 0.19 and `transformers` 4.57 broke the import path; we patched it at load time. An NCCL timeout also silently hung training for ≈ 28 hours, so we added a watchdog that kills the process after 15 minutes with no step-advance and resumes from the latest checkpoint.

Final pipeline and a bug we caught. The submitted pipeline is self-consistency $k = 3$ + the GRPO adapter, on all 943 private items. The driver writes one JSONL record per item so it can resume, then converts to the submission CSV. On the first full run, 10 items came back empty: long reasoning traces filled the 24,576-token context window and there was zero budget left for new tokens, which vLLM rejects. We fixed `run_inference.py` to clamp `max_tokens` to the remaining context budget (and skip the item if fewer than 256 tokens are left) and re-ran those 10 items. The final CSV has 943 rows, 0 empty, 0 errors, and scored **0.699** on the public board (rank 42/105) and

0.627 on the private board (rank 27/105). We also tried a longer-reasoning variant (`max_tokens = 16,384`, `max_model_len = 49,152`); it scored 0.692 public / 0.612 private on the leaderboards, slightly worse, so we did not adopt it. Qwen3-4B-Thinking-2507 supports 262,144 tokens natively, so KV cache was not the binding constraint; the free-form ceiling we are hitting is a reasoning-quality limit, not a token-budget limit.

4 Experiments

4.1 Baselines

Our baseline is the milestone pipeline: Qwen3-4B-Thinking-2507 under INT8 on a single GPU, the grader-aware system prompt, and the Qwen-recommended sampling defaults. That baseline scored 0.515 public / 0.459 private on the Kaggle leaderboards, and every later number we report is a delta on top of that. We do not include simpler models like a multi-layer perceptron or linear regression as baselines, because they cannot do free-form math reasoning over natural-language problem statements in any meaningful way. Comparing additive method changes to the same backbone is what made sense at this stage.

4.2 Evaluation

We score with the provided judge: `Judge.auto_judge` for free-form items, exact-letter match for MCQ items. For fast method development we use the 44-item eval slice from Section 3 (10 MCQ, 34 free-form). Kaggle has two boards. The public board scores submissions on all 943 items of `private.jsonl` — this is what we watched during the competition. The private board scores on a separate hidden subset we never see, revealed only at the deadline to catch teams that overfit `private.jsonl`.

4.3 Implementation details

- **Hardware (final):** $2 \times$ NVIDIA A40 (46 GB each); vLLM tensor-parallel = 2; BF16 weights. The milestone ran on one GPU under INT8 (BitsAndBytes); we switched because Qwen3-4B is only ≈ 8 GB in BF16 (quantization was mostly dequant overhead), and the move to BF16 with TP= 2 raised throughput from ≈ 175 to ≈ 786 output tokens/second.
- **Wall-clock:** a full 943-item private run at SC $k = 3$ takes about 4 days (≈ 10 items/hour); GRPO training was ≈ 14 hours (1 epoch, 561 steps).
- **Model:** Qwen/Qwen3-4B-Thinking-2507 through vLLM with `dtype="bfloat16"`, `enable_prefix_caching=False`, `gpu_memory_utilization=0.75`, `max_model_len=24,576`, `max_num_seqs=128`, `max_num_batched_tokens=16,384`, `tensor_parallel_size=2`, and `enable_lora=True` when the GRPO adapter is active.
- **Sampling (final):** `temperature=0.6`, `top_p=0.95`, `top_k=20`, `min_p=0.0`, `repetition_penalty=1.0`.
- **Self-consistency:** $k = 3$ in the final pipeline ($k = 8$ tested and found neutral).
- **Reproducibility:** all of the above are locked in `run_inference.py`. `run_inference()` loads the base model, downloads the LoRA adapter from `k1mittal/cse151b-grpo-lora` if not local, runs the pipeline on `data/private.jsonl`, and writes `csv/submission.csv`.

4.4 Results

Result 1: $0.515 \rightarrow 0.699$ public and $0.459 \rightarrow 0.627$ private on the Kaggle leaderboard. Our submitted pipeline scored **0.699** on the public board (rank 42/105) and **0.627** on the private board (rank 27/105): $+0.184/+0.168$ absolute and $+36\%/+37\%$ relative over the milestone. Rank rose from 42 public to 27 private, a $+15$ -place jump — if the layered pipeline had overfit `private.jsonl` we would expect rank to drop on the hidden subset, not improve. All 943 private items came back with 0 empty rows and 0 errors once we caught the context-budget bug in Section 3. We also submitted a second pipeline as a sanity check (longer reasoning budget, everything else fixed) which scored 0.692 / 0.612, slightly below the adopted run on both boards; the full comparison is in the reasoning-budget ablation, Section 4.5. Figure 3 shows the milestone-to-final progression on the private board.

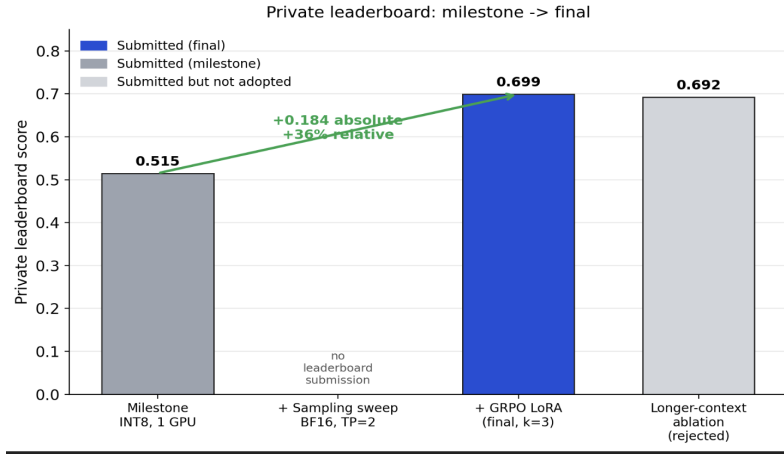


Figure 3: Private-leaderboard progression from the milestone submission (INT8, 1 GPU; 0.515 public / 0.459 private) to the final submission (BF16 + TP= 2 + sweep + SC $k=3$ + GRPO LoRA; 0.699 / 0.627), +0.184 absolute on the public board (+36% relative). The longer-context ablation submission (0.692 / 0.612) is included for reference but was not adopted.

Result 2: $\approx 4.5\times$ inference throughput from INT8 ($1\times$ GPU) to BF16 + vLLM TP= 2 ($2\times$ A40). The milestone pipeline ran Qwen3-4B-Thinking-2507 under INT8 (BitsAndBytes) on a single GPU at ≈ 175 output tokens/second. Switching to BF16 weights with vLLM tensor-parallel = 2 on $2\times$ A40 raised throughput to ≈ 786 tokens/second, roughly $4.5\times$. The reason quantization was a loss rather than a win is that Qwen3-4B is only ≈ 8 GB in BF16 — well within a single A40’s 46 GB — so the INT8 path spent most of its budget on dequantization overhead with no memory headroom to gain. This change is what made the rest of the final phase tractable: at the milestone’s ≈ 175 tok/s, a single full-private run at $k = 3$ would have taken ≈ 17 days instead of the ≈ 4 it actually took, and we would not have had wall-clock to fit the sampling sweep, the SC study, and a GRPO training run on top of it. Figure 4 shows the two operating points.

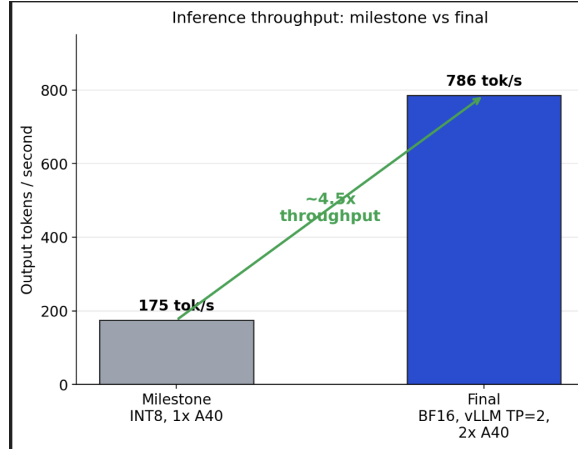


Figure 4: Output throughput at the milestone (INT8, single A40) vs. the final pipeline (BF16, vLLM tensor-parallel = 2 on $2\times$ A40), measured during full-private runs. The $\approx 4.5\times$ speedup is what made the 24-config sweep, the SC study, and one full epoch of GRPO fit in our final-phase wall-clock budget.

4.5 Ablations

We isolate the contribution of each component of the final pipeline by removing or replacing it while holding everything else fixed. All numbers below are on the 44-item eval slice from Section 3 (10

MCQ, 34 free-form), with the same sampling settings, seeds, and prompt across rows. Table 1 gives the layer-by-layer breakdown; the paragraphs after it isolate each layer in more detail.

Table 1: Layer-by-layer ablation of the final pipeline on the 44-item eval slice. All rows use Qwen3-4B-Thinking-2507 under BF16 + vLLM TP= 2 with (T=0.6, top_p=0.95, top_k=20, min_p=0). Each row adds one layer to the row above.

Pipeline stage	MCQ ($n = 10$)	Free-form ($n = 34$)	Overall ($n = 44$)
Sampling sweep (best)	80.0%	64.7%	68.2%
+ Self-consistency ($k = 8$)	90.0%	64.7%	70.5%
+ GRPO LoRA	90.0%	64.7%	70.5%

Self-consistency size ablation. At $k = 8$ the vote scored 70.45% overall, identical to the best single sample on free-form, with a 1-item MCQ swing (8/10 $\rightarrow$ 9/10) that we treat as noise on a 10-item split. Items the model gets right were already high-confidence; items it gets wrong tended to be wrong consistently across rollouts. We dropped k from 8 to 3 in the final pipeline — about $2.7\times$ cheaper at the same expected accuracy on this slice — and kept SC as a cheap robustness check rather than a measured gain. The flatness should be read as “no signal we can resolve on a slice this small,” not “we proved SC does nothing.”

GRPO adapter ablation. On the eval slice the GRPO adapter scored 70.45%, identical to SC alone. Probable causes are listed in Section 3: one low-learning-rate epoch, group size 4 instead of the 8–16 from the published recipes, completions truncated at 2,048 tokens, and a sparse 0/1 reward. We kept the adapter in the final pipeline because validation showed no regression on either the eval slice or the Kaggle leaderboards.

Reasoning-budget ablation. We doubled the context budget (`max_tokens` = 16,384, `max_model_len` = 49,152) and re-ran the final pipeline end-to-end, leaving everything else fixed. The variant scored 0.692 public / 0.612 private — -0.007 on the public board against the adopted submission’s 0.699. Both setups use less than 20% of Qwen3-4B-Thinking-2507’s 262,144-token native window, so the KV cache was not the binding constraint. The wall we are hitting is reasoning quality, not tokens.

5 Discussion

We started the final phase from 0.515 public / 0.459 private and ended at 0.699 public (rank 42/105) / 0.627 private (rank 27/105), $+36\%/+37\%$ relative. The rest of this section covers what worked, what did not, and what we would try next.

Achievements. The pipeline runs end-to-end from a single function call (`run_inference()` in `run_inference.py`). The LoRA adapter is on HuggingFace Hub at `k1mittal/cse151b-grpo-lora`, so verification does not need any local model files. The full 943-item private run completed with zero empty or errored items after we caught and fixed the context-budget bug in Section 3. On the eval slice the model’s strongest split is MCQ, where the SC + GRPO pipeline reaches 90% (Table 1).

Bottlenecks and Mitigation. The sampling sweep was the largest single contribution in the final phase; SC at $k = 8$ and the GRPO adapter were both flat on the eval slice and we kept them for robustness. The persistent bottleneck is free-form accuracy, which stayed near 65% through every layer. Inside the free-form split, multi-answer items (questions wanting a list of values rather than a single value) were the weakest subset across every method we tried; our general prompt was not specialized for those items. The reasoning-budget ablation (Section 4.5) rules out token budget as the cause: the wall is reasoning quality, not tokens. We also did not get through every idea on our list because of time constraints. From the milestone plan we deviated in three places: we moved inference from INT8 on one GPU to BF16 with vLLM TP= 2 for throughput; we tested SC at $k = 8$ rather than the planned $k \in \{3, 5\}$, found no gain, and shipped $k = 3$ for cost; and GRPO ran with reduced group size and shorter completions because of single-box memory.

Lessons Learned.

- **Run small eval chunks before full runs.** Running each change on a small chunk of the public set first gave us a clean read on whether the change was actually helping the model, without paying for a full-private run on every experiment.
- **Use overnight runs to parallelize work.** Wrapping the notebook in bash scripts and letting it run overnight on the remote server meant we could keep experiments going in the background while we worked on other things, which gave us back a lot of wall-clock time.
- **If given more time.** (i) Use the model’s context tokens better and write prompts that target the multi-select questions specifically, where our general prompt left accuracy on the table; (ii) augment GRPO fine-tuning with NVIDIA’s Nemotron dataset to strengthen the post-training stage.

Team Responsibilities.

- **Ketan Mittal** — grader analysis, prompt design, sampling-parameter sweep, final private run, bug-fixes, and the submission pipeline.
- **Anirudh Annabathula** — inference infrastructure (vLLM TP= 2, BF16 on $2 \times A40$), GRPO training setup on the GPUs, and the presentation.
- **Kaii Bijlani** — dataset analysis, the Claude-based answer filter over the public set, and the presentation.

LLM Usage

We used an LLM assistant (Anthropic Claude) for two things in the report-writing process: (i) help with wording, mostly tightening up phrasing in this report and in our system prompts, and (ii) help making sense of hard-to-read error messages in LaTeX and Python while we were writing the report and building the pipeline. We also used Claude as a *verifier* for the GRPO training set — it read each public-set item and flagged ones whose published gold answer was wrong or ambiguous — which is methodological use rather than report-writing assistance, and is described as such in Section 3. All the technical decisions, code, experimental runs, and numbers in this report are ours.

References

- [1] Qwen Team. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025. URL <https://arxiv.org/abs/2505.09388>.
- [2] Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Mingchuan Zhang, Y. K. Li, Y. Wu, and Daya Guo. DeepSeekMath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024. URL <https://arxiv.org/abs/2402.03300>.
- [3] An Yang et al. Qwen2.5-Math technical report: Toward mathematical expert model via self-improvement. *arXiv preprint arXiv:2409.12122*, 2024. URL <https://arxiv.org/abs/2409.12122>.